

Solution exercise set 6

Problem 0:

a) Using the standard estimators $\hat{\mu} = \bar{X}$ and $\hat{\sigma} = S = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2}$ we obtain the estimates:

$$\begin{aligned}\hat{\mu} &= 175.8 \\ \hat{\sigma} &= 7.66\end{aligned}$$

Furthermore, we know that $\hat{\mu}$ is unbiased since

$$E(\hat{\mu}) = E(\bar{X}) = \frac{1}{n} \sum_{i=1}^n E(X_i) = \frac{1}{n} \sum_{i=1}^n \mu = \mu$$

Lastly, we have that (by independence)

$$SD(\hat{\mu}) = \sqrt{Var(\bar{X})} = \sigma/\sqrt{n}$$

which yields that

$$SD(\hat{\mu}) = \hat{\sigma}/\sqrt{n} = 7.66/\sqrt{10} = 2.42.$$

b) In your introductory course to statistics you have learned that a $(1 - \alpha)100\%$ confidence interval for μ in the normal distribution (when σ^2 is unknown) is given by:

$$[\bar{X} \pm t_{\alpha/2, n-1} \frac{S}{\sqrt{n}}]$$

If we want a 95% conf. int. we let $\alpha = 0.05$ which gives $t_{\alpha/2, n-1} = t_{0.025, 9} = -2.262$, and a 95% conf. int. for μ becomes:

$$\begin{aligned}& [175.8 - 2.262 \cdot \frac{7.66}{\sqrt{10}}, 175.8 + 2.262 \cdot \frac{7.66}{\sqrt{10}}] \\ & = [170.3, 181.3]\end{aligned}$$

You may (or may not) also have learned that we can obtain a CI for σ by $\left[\sqrt{\frac{(n-1)S^2}{\chi_{1-\alpha/2, n-1}^2}}, \sqrt{\frac{(n-1)S^2}{\chi_{\alpha/2, n-1}^2}} \right]$.

With $\alpha = 0.05$ we have $\chi_{0.975, 9}^2 = 19.023$ and $\chi_{0.025, 9}^2 = 2.700$, and we get 95% conf. int.:

$$\left[\frac{9 \cdot 7.66^2}{19.023}, \frac{9 \cdot 7.66^2}{2.700} \right] = [5.27, 13.98]$$

We can thus summarize the results for the analysis of the hardness data as:

$$\begin{aligned}\hat{\mu} &= 175.8 \quad \text{with 95\% c.i. } [170.3, 181.3] \\ \hat{\sigma} &= 7.66 \quad \text{with 95\% c.i. } [5.27, 13.98]\end{aligned}$$

c) See R code: `bootstrap_examples.R`.

d) See R script file.

The remaining problems are solved using R only